

Hedge Portfolio · v2 Pipeline Draft

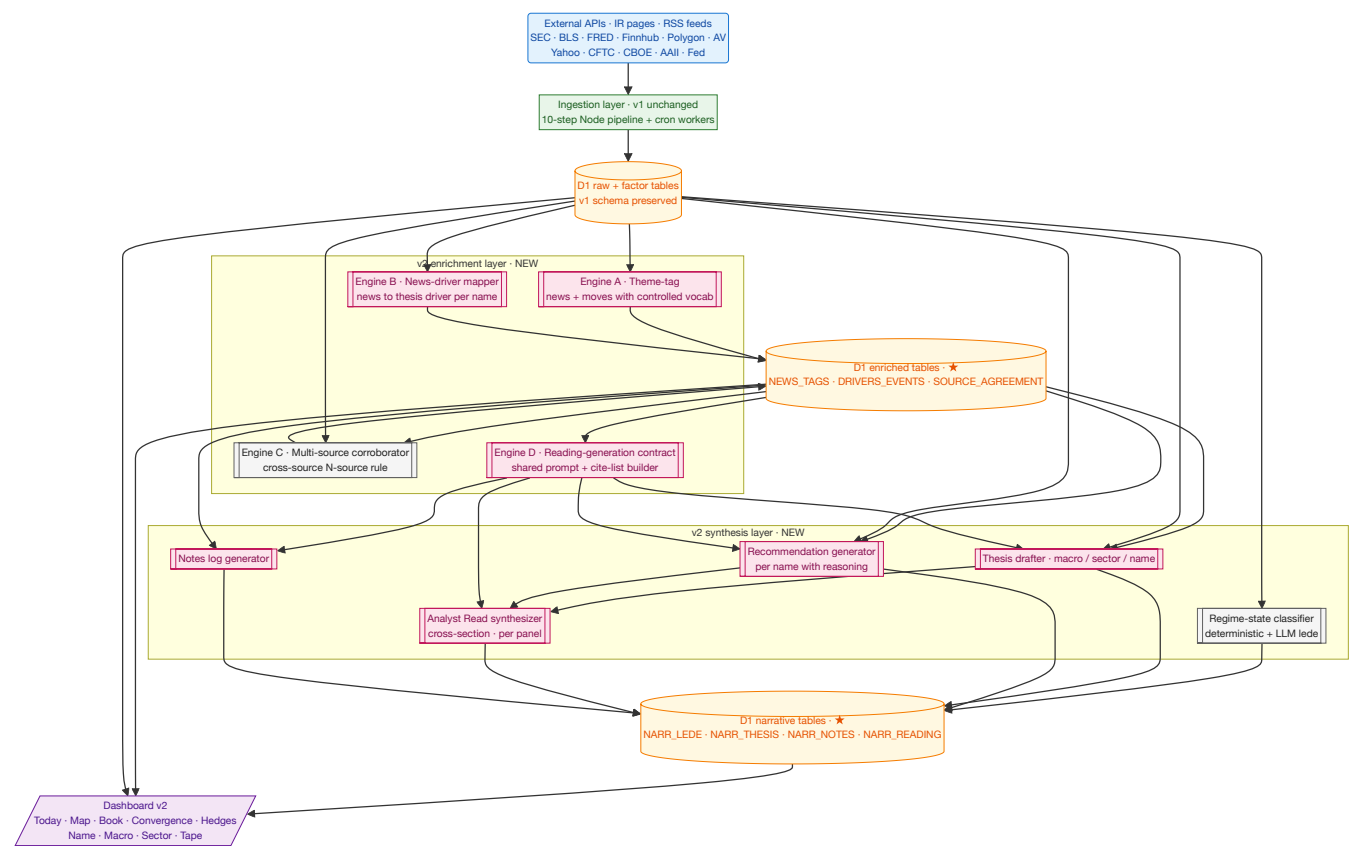
Diagrams + tables for the data + processing pipeline that would be required to power the v2 dashboard mockup (`dashboard/mockup/index.html`). This is a proposal — no schema, worker boundaries, or prompts are committed yet. The single most important difference from v1 is that **news flows directly into conclusions** for macro, sector, and per-name readings.

Diagram conventions

Tag prefix	Shape	Layer	Meaning
R<n>	rounded (…)	source	external data source or static config
C<n>	rectangle [...]	code module	ingestion / fetcher / orchestrator
T<n>	cylinder [(…)]	D1 table	persisted state — ★ marks new in v2
A<n>	hexagon {{…}}	LLM agent	prompt-based step
A<n>	subroutine [[…]]	deterministic agent	SQL / numeric / rule-based step
E<n>	subroutine [[…]]	engine reuse	call into a cross-cutting engine (Part 2)
D<n>	parallelogram [/…/]	dashboard field	rendered element on the v2 mockup

Tag numbering resets per cluster.

Part 1 · System overview

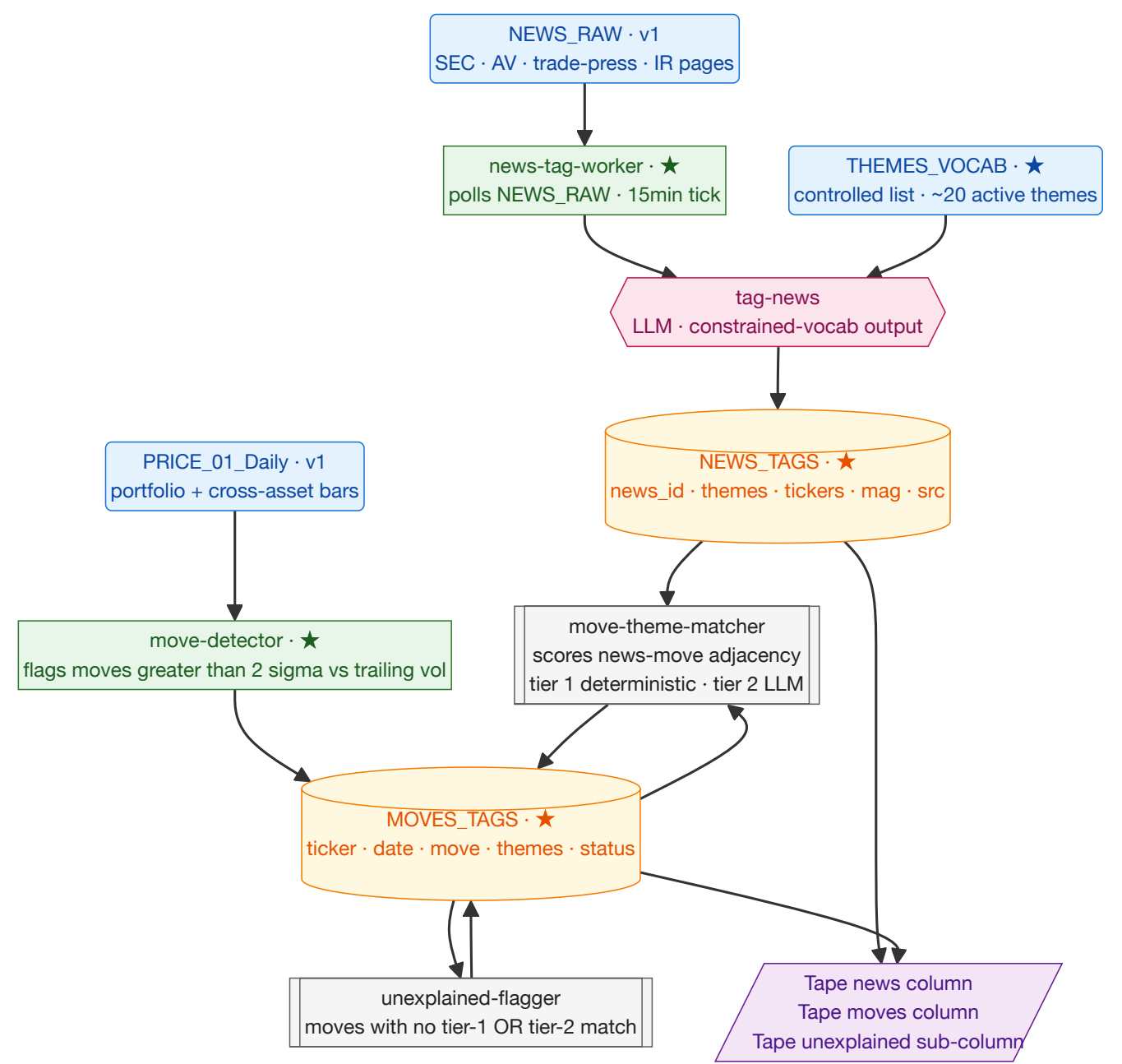


Layer	v1 today	v2 adds
Ingestion	10-step Node + cron workers	unchanged
Raw + factor	~40 D1 tables	unchanged
Enrichment	Per-name news funnel · macro-intel summary	Theme-tag · News-driver mapper · Multi-source corroborator
Synthesis	Per-entity narrator (3-stage)	Reading-engine contract · Thesis drafter · Recommendation · Notes · Read
Narrative tables	NARRATIVE_01_Content	NARR_LEDE · NARR_THESIS · NARR_NOTES · NARR_READING (5 reading types)

Part 2 · Cross-cutting engines

Engine A · Theme-tag · news + moves with controlled vocabulary

Foundation for the Tape and for theme-aware readings. Every news item and every >2σ move gets tagged from a small fixed vocabulary so colors stay scannable.



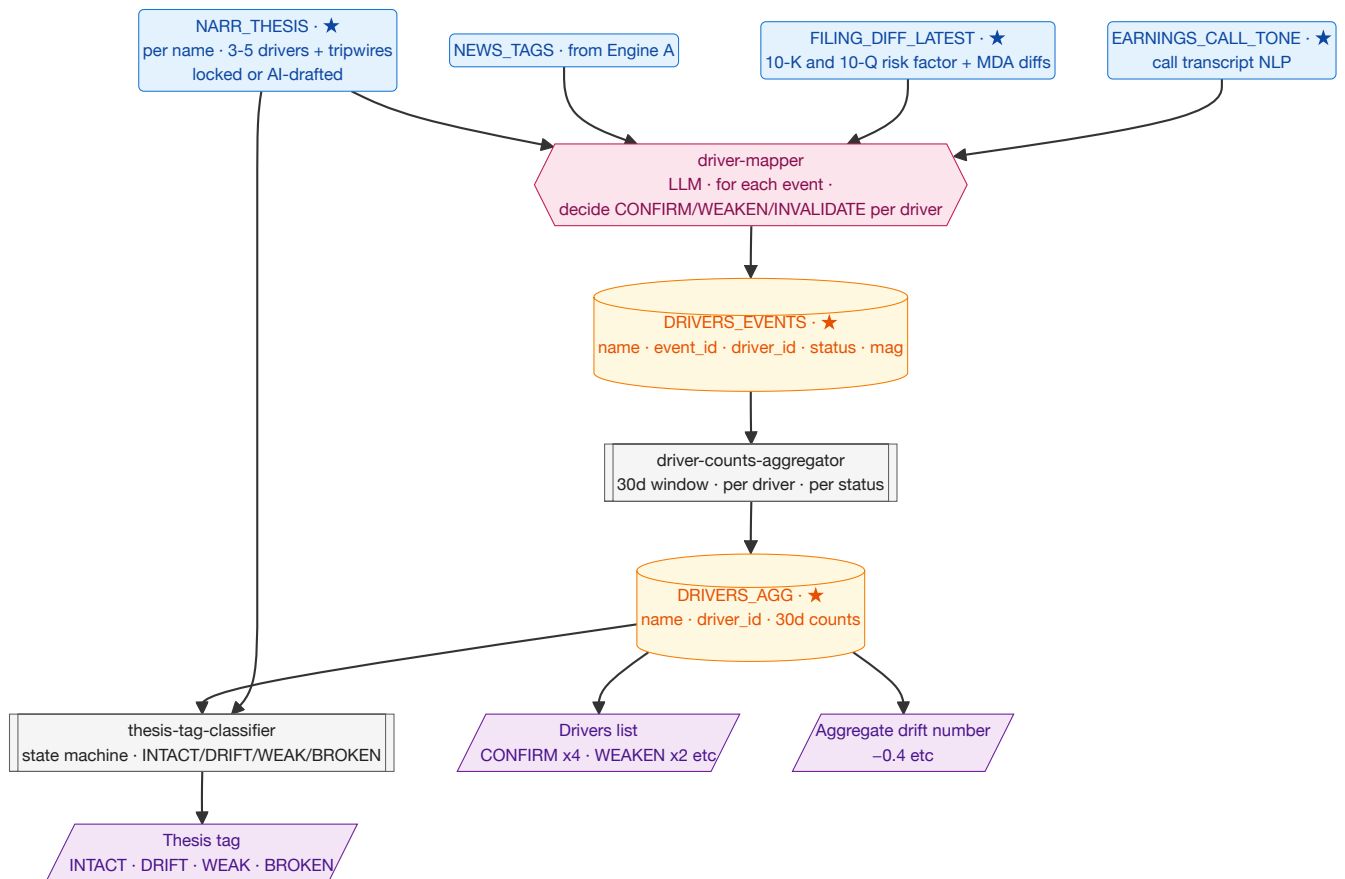
Tag	Notes
R3	Hand-curated, ~20 themes max, versioned. Examples: tariffs , oil , ai-capex , rates , china , earnings , ev-transition , geopolitics .

Tag	Notes
A1	LLM tag step. Output: { themes, tickers, mag, dir }. Constrained vocabulary — themes outside R3 are rejected.
A2	Tier 1 (deterministic): for each move, find news in prior 14d touching same ticker with mag > 0.3. Tier 2 (LLM): if Tier 1 finds nothing, propose a likely theme.
A3	A move is unexplained when no Tier-1 OR Tier-2 match exists. Surfaced, not hidden.
T1	The foundation table for the Tape and for all news-aware readings.
T2	One row per $>2\sigma$ move. Status: tagged-tier-1 / tagged-tier-2 / unexplained .

Inputs	Processing	Outputs (mockup fields)
News raw text	LLM tag (constrained vocab)	NEWS_TAGS rows
Price daily bars + vol	Move detector	MOVES_TAGS rows
Theme vocabulary	Tier 1 + Tier 2 matcher	Tape news + moves columns
	Unexplained flagger	Tape unexplained sub-column

Engine B · News-driver mapper · news → thesis driver per name

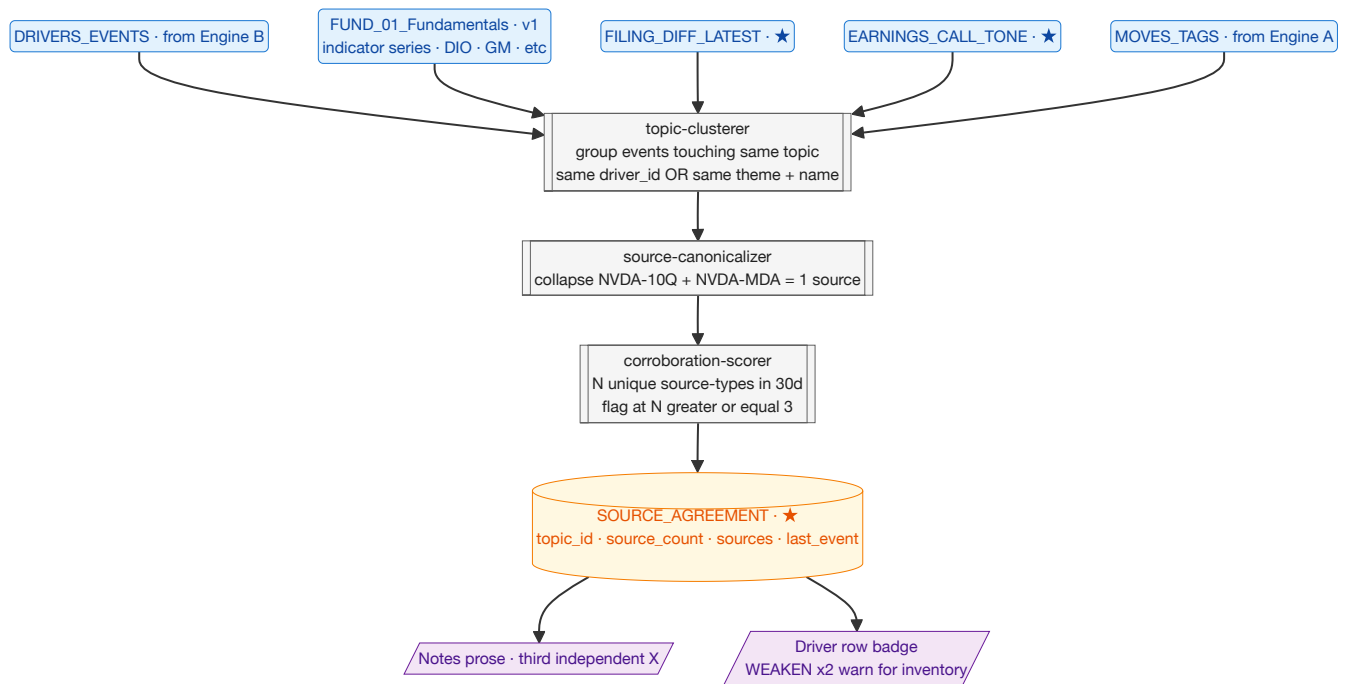
Makes claims like *"DC growth confirming +4 events"* or *"third independent inventory signal"* into deterministic counts rather than LLM imagination.



Tag	Notes
A1	Inputs: locked driver list of the name + the news/filing/call event. Output: { driver_id, status, mag, evidence_quote }. Items not mapping to any driver are dropped.
A2	Deterministic 30-day rolling aggregator. Produces the CONFIRM ×4 / WEAKEN ×2 counts.
A3	Rule-based state machine. Any INVALIDATE → BROKEN ; WEAKEN with no offsetting CONFIRMS → WEAK ; ≥ 2 drivers softening → DRIFT ; else INTACT .
T1	One row per (name, event, driver). Stores the evidence quote so the dashboard can show <i>why</i> a count moved.

Engine C · Multi-source corroborator · the N-source rule

A note that says "third independent signal" is only honest if the system actually checks across sources. Pure-deterministic guard against fabrication.

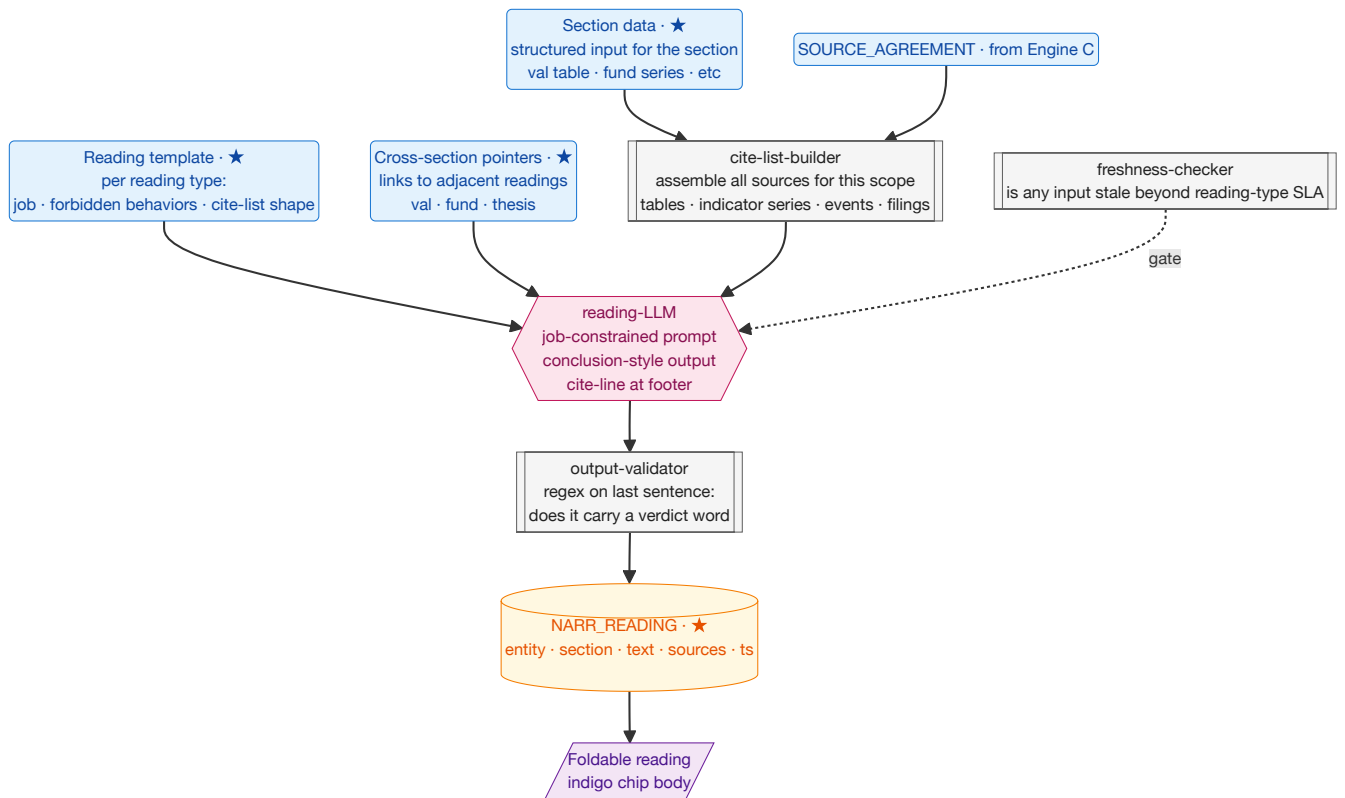


Tag	Notes
A1	Joins items across <code>DRIVERS_EVENTS</code> , <code>FUND_01</code> , <code>FILING_DIFF_LATEST</code> , <code>EARNINGS_CALL_TONE</code> , <code>MOVES_TAGS</code> .
A2	Filings + their MDA passages = one source; trade press from the same outlet across days = one source. Prevents N-count over-counting tightly correlated reports.
A3	Counts unique canonical sources per topic over 30d. Topics with $N \geq 3$ are eligible for "multi-source" claims in readings.

The corroboration is computed *before* the synthesis layer. Synthesis prompts are *given* the topic + source list and only write prose around it.

Engine D · Reading-generation contract · shared prompt + cite-list builder

Every "reading" in the v2 mockup (per-section + Analyst Read lede) is produced by this engine. Contract enforces the writing principles in `DASHBOARD_AI_INTEGRATION.md` Section 16.



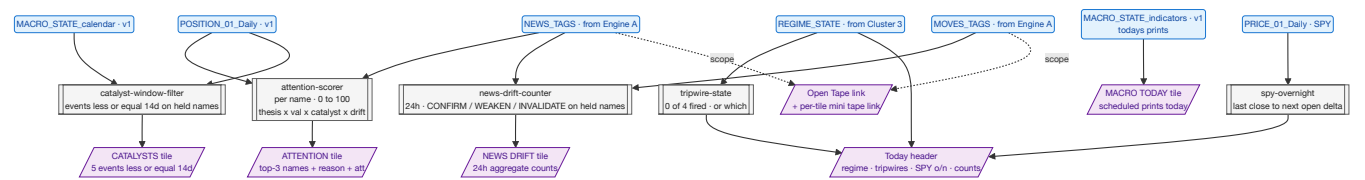
Reading template fields per reading type:

Template field	Example for Fundamentals reading
Job	Land on a company verdict — peaking / accelerating / harvesting / transitioning / declining
Must end on	A verdict label
Forbidden	Listing comma-separated values · Wall Street metaphors · uncited claims
Required cross-refs	Thesis driver-N · Valuation reading
Cite-list shape	20q series IDs + composite scores + cross-references
Output validator	Last sentence must contain one of: <i>peaking, accelerating, harvesting, intact, watch, etc.</i>

Part 3 · Surface clusters

Cluster 1 · Today notification surface

Notification only — no AI essay. Every tile is a deterministic count or list; clicking opens the Tape pre-filtered.

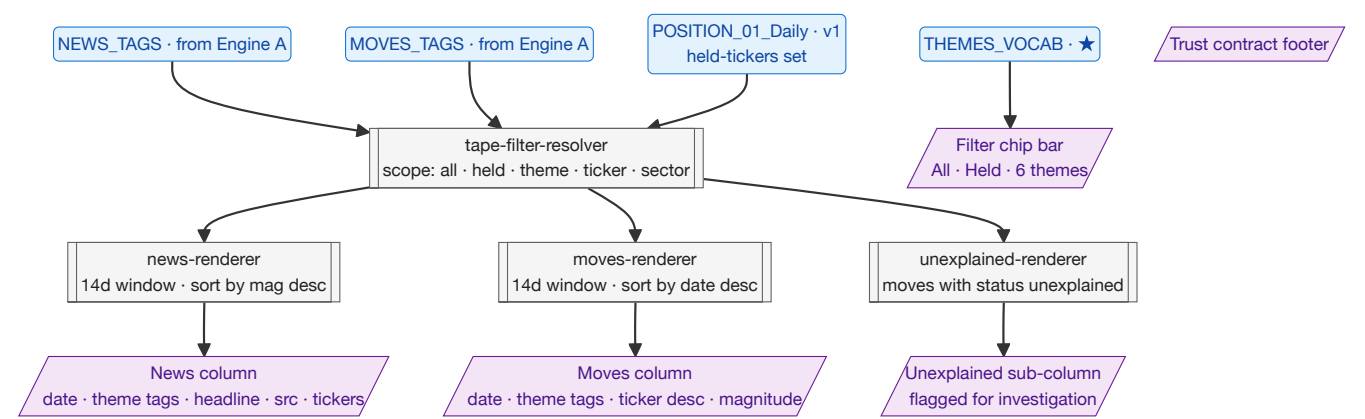


Inputs	Processing	Outputs
Position book · thesis · val z · drift	Attention scorer	ATTENTION tile rows
Calendar · held tickers	Catalyst window filter	CATALYSTS tile rows
Macro release schedule for today	Direct fetch	MACRO TODAY tile rows
Tagged news + moves over 24h on held names	News-drift counter	NEWS DRIFT counts
Regime + tripwires + SPY o/n	Tripwire monitor · SPY o/n calc	Today header state strip
Theme + ticker scope per tile	(passthrough)	Per-tile + global Tape links

The only AI involvement is upstream (Engine A's tagging). The Today surface itself is deterministic and re-renders on every page load.

Cluster 2 · Tape slide-out

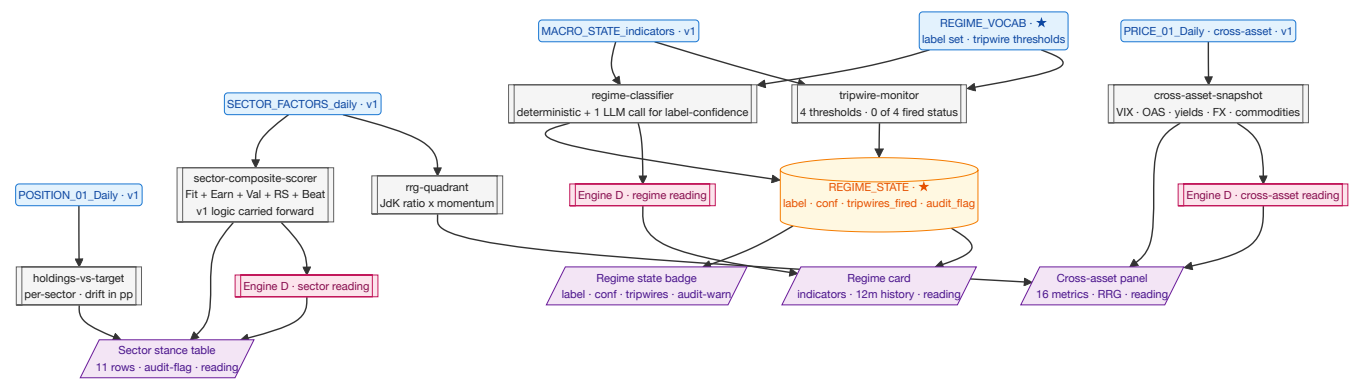
Pure presentation of Engine A output, with filter chips on top. **No causal AI text in this surface.**



Inputs	Processing	Outputs
Tagged news + moves (14d)	Filter resolver · scope-aware	News + Moves columns
Held-tickers set	Held-only sub-filter	
Theme vocabulary	Filter chip rendering	Filter chip bar
Move status (unexplained)	Unexplained renderer	Unexplained moves sub-column

Cluster 3 · Map · regime state badge + sector + cross-asset

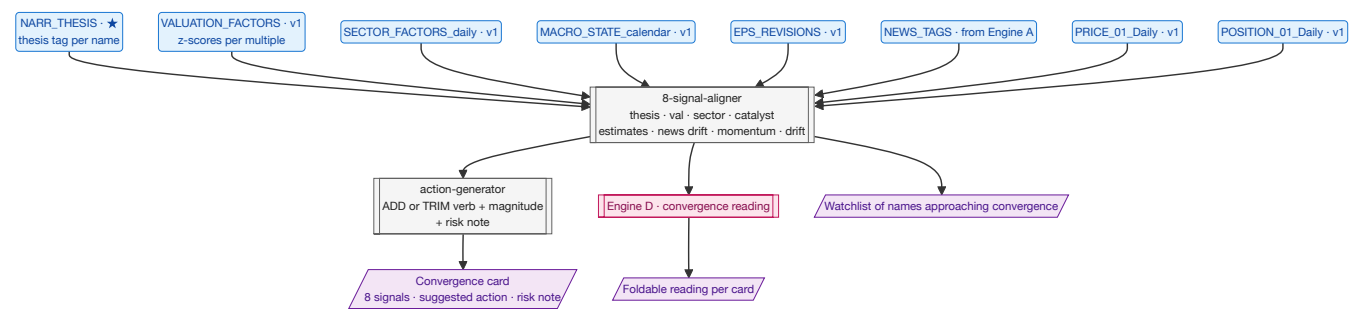
Big shift from v1: the MAP READ essay is gone; in its place a single-line **regime state badge** with all-deterministic state. Per-panel readings stay (foldable, Engine D).



Inputs	Processing	Outputs
12 macro indicators · vocab	Regime classifier · tripwire monitor	Regime state badge · Regime card
Sector factors · book	Composite scorer · holdings-vs-target	Sector stance table · audit-flag
Cross-asset prices	Snapshot · RRG quadrant	Cross-asset panel · RRG note
Engine D	Per-panel reading generation	3 foldable readings

Cluster 4 · Convergence

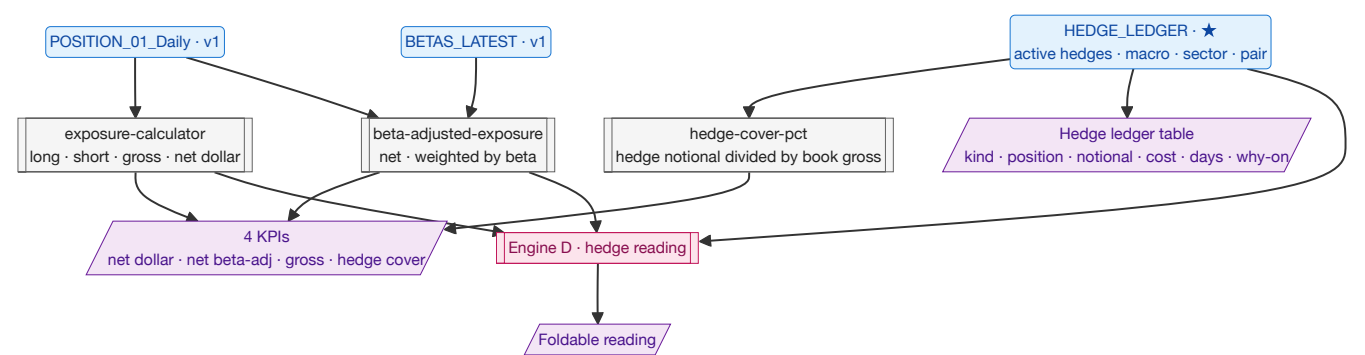
8-signal alignment per name. Action verb when ≥ 3 align same direction.



Inputs	Processing	Outputs
8 signal sources	Aligner · firing-count	Card signal list
Aligner output	Action generator	Verb · magnitude · risk note
Engine D	Convergence reading	Foldable per-card reading
Aligner output (≥ 2 of 8)	Watchlist filter	Approaching-convergence footer

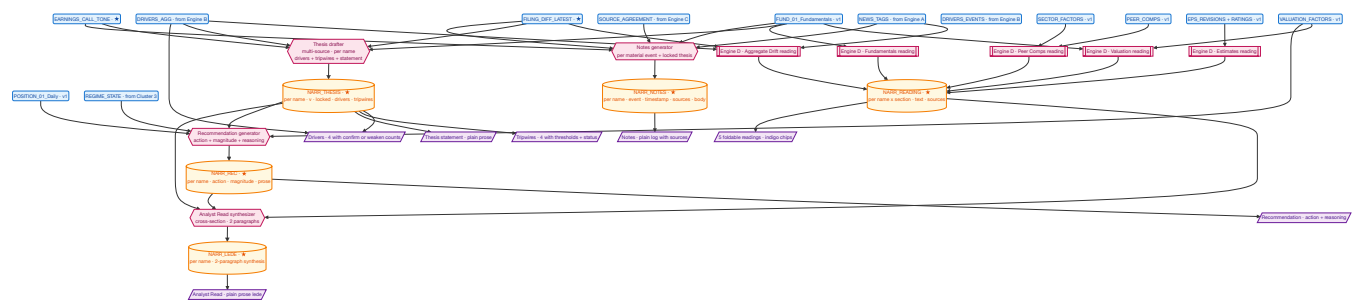
Cluster 5 · Hedges

Position aggregation + beta-adjusted exposure. Light surface.



Cluster 6 · Name slide-out

The biggest surface — and where the news-into-conclusions wiring matters most. Leade + 5 per-section readings + thesis + recommendation + notes.



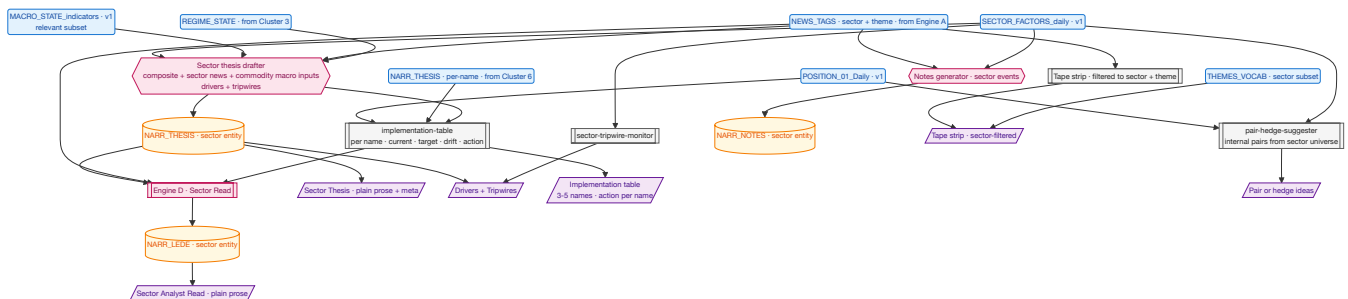
Output (mockup field)	Inputs	Processing
Analyst Read (plain lede)	All per-section readings · thesis · recommendation	A4 cross-section synthesizer
Thesis statement (plain)	Driver counts · filing diffs · call tone · fundamentals	A1 Thesis drafter
Drivers (CONFIRM ×N etc.)	DRIVERS_AGG (deterministic counts)	Engine B counts

Output (mockup field)	Inputs	Processing
Macro Thesis	Indicators · FOMC · regime state · macro-themed news	A1 Macro thesis drafter
Drivers + Tripwires	Threshold table + current values	A2 deterministic monitor
Signposts (next 30d)	Forward calendar + indicator thresholds	A3 signpost builder
Positioning implication	Thesis + position book + hedge ledger	A4 Macro recommendation
Notes (plain log)	Macro-themed news + indicator prints	A5 Notes generator

News dependency: macro thesis claims like *"stagflation-light flavor strengthening"* or *"HY OAS tightened — single largest piece of evidence the regime is not the front-edge of a recession"* are interpretations of indicator series in light of macro news. Without `NEWS_TAGS` filtered to macro themes, the thesis drafter has no way to surface "FOMC speakers more hawkish" or "tariff escalation" as inputs.

Cluster 8 · Sector slide-out

Same structural treatment as macro, per sector. Drivers/tripwires are sector-specific (Energy: WTI, OPEC discipline, US rig count, EM demand). Implementation table maps thesis → per-name actions.



Output (mockup field)	Inputs	Processing
Tape strip · sector-filtered	Sector tickers + sector themes	Engine A passthrough
Sector Analyst Read	Thesis · implementation · per-name convergence cross-ref	Engine D (Sector Read)
Sector Thesis	Sector composite · macro relevant subset · sector-themed news	A1 Sector thesis drafter
Drivers + Tripwires	Per-sector thresholds + current values	A2 deterministic monitor
Implementation table	Per-name positions + per-name thesis state	A3 deterministic table builder
Pair / hedge ideas	Sector universe + book composition	A4 deterministic suggester

News dependency: the Energy sector thesis claim that *"OPEC+ discipline + capital-return-focused US shale management together support a higher floor under WTI than consensus models"* requires news — OPEC+ outcomes are filings, US shale capex discipline is news + earnings calls, EM stockpiling is news. Without sector-themed news in `NEWS_TAGS` , the thesis collapses to "WTI is at \$78".

Part 4 · Pipeline tree summary

